Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

Abstract

Conformal prediction provides distribution-free coverage guarantees, but these degrade under distribution shift—and practitioners lack tools to anticipate *which* deployed models will fail before deployment. We propose SHAP concentration—the fraction of feature importance concentrated in the top feature—as a **pre-deployment diagnostic** for conformal prediction vulnerability. Using COVID-19 as a case study across 8 supply chain tasks experiencing identical temporal shift yet coverage drops ranging from 0.1% to 77.1% (all paired $p \leq 0.005$, 50 seeds), we show that, for gradient-boosted models, SHAP concentration is strongly associated with failure severity: $\rho = 0.853, p < 0.001$ across 16 multiclass tasks in 9 domains (bootstrap 95% CI $[0.50, 0.96]$; Kendall $\tau = 0.667$, $p < 0.001$). Standard shift detectors (MMD, C2ST) detect shift for all tasks uniformly but do not distinguish catastrophic from robust outcomes ($\rho \leq 0.19$, all $p > 0.6$); among evaluated pre-deployment diagnostics, SHAP concentration shows the strongest association with failure severity. External validation across 9 non-supply-chain datasets shows partial transfer with asymmetric evidence: Covertype ($C = 49.8\%$, 81.8 pp drop) is deterministic across 10 seeds, most low-concentration datasets are coverage-robust, Shuttle is near-deterministic (9/10), KDDCup99 is a seed-dependent intermediate case, and Stack Overflow ($K = 3$) exhibits a near-binary ceiling effect and is excluded from the multiclass primary endpoint ($n = 16$). We prove a formal score-inflation theorem under explicit assumptions showing that APS conformity-score bounds worsen monotonically with concentration, verified on all applicable tasks. A decision framework operationalizes the diagnostic as an exploratory rule: compute SHAP concentration pre-deployment; use a conservative uncertainty band around the 40% threshold; suggest that quarterly retraining may partially recover coverage when vulnerable (+19 pp, $p = 0.04$ unadjusted; Holm-corrected $p = 0.12$); and defer routine retraining for lower-risk tasks.

Submitted to the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026). **To be used for reviewing only.**

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience identical temporal shift yet exhibit coverage drops ranging from negligible (<1%) to catastrophic (>70%).

Our contributions:

1. **Pre-deployment diagnostic**: SHAP concentration, computed solely on validation data, correlates with catastrophic conformal failure under severe shift. Across 16 multiclass tasks in 9 domains: $\rho = 0.853$, $p < 0.001$ (Kendall $\tau = 0.667$, $p < 0.001$; bootstrap 95% CI $[0.50, 0.96]$). Within supply chain alone: $\rho = 0.833$, $p = 0.010$ ($n = 8$). SHAP concentration achieves stronger correlation than alternative pre-deployment diagnostics—native feature importance ($\rho = 0.667$, $p = 0.071$) and ensemble disagreement ($\rho = 0.452$, $p = 0.260$)—though the difference is not formally testable at $n = 8$. Binary APS exhibits a structural ceiling effect; concentration is diagnostic specifically for multiclass settings.
2. **Formal theory**: Theorem 1 establishes that, under an additive feature-decomposition model, APS score and coverage bounds worsen monotonically with concentration parameter C (Section 4). The bound is verified on all 5 applicable tasks. We confirm empirically that test conformity scores stochastically dominate calibration scores for catastrophic tasks (KS = 0.68–0.96, all $p < 10^{-10}$; Figure 3). Catastrophic tasks show *decreasing* prediction entropy—the opposite of standard intuition—because the model misclassifies confidently rather than randomly.
3. **Quantification with statistical rigor**: Coverage drops across 8 tasks range from 0.1% to 77.1%, all with paired $p \leq 0.005$ (Wilcoxon, 50 seeds). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though significance is not maintained for 2 of 8 jackknife samples due to reduced power at $n = 7$.
4. **Operational framework (exploratory)**: Quarterly retraining can partially mitigate vulnerable task coverage loss by +19 pp ($p = 0.04$); robust tasks (99.8% coverage) show little intervention benefit in this setting. Under the at-risk criterion (drop >15 pp), the framework shows a precision–recall trade-off across 30–45% thresholds (precision 0.71→1.00; recall 0.83; Appendix D).
5. **Binary ceiling effect**: Binary APS prediction sets ($\{0\}, \{1\}$, or $\{0, 1\}$) are structurally protected against the coverage degradation that concentration detects. This boundary condition clarifies when the diagnostic applies (multiclass) and when it does not (binary).
6. **External domain validation**: The SALT-derived 40% threshold is partially transferable across 9 external non-supply-chain datasets without tuning. Ten-seed replication yields deterministic/near-deterministic behavior for 7/9 datasets (6 deterministic, 1 near-deterministic): Forest Covertype is correctly flagged as catastrophic ($C = 49.8\%$, 10/10 seeds), while KDDCup99 ($C = 21.1\%$, drop $= 15.9 \pm 21.4$ pp) and Stack Overflow (near-binary ceiling) are boundary cases that require caveats; external catastrophic evidence is concentrated in Covertype.
7. **Shift detection $\neq$ severity prediction**: Standard shift indicators (MMD, C2ST, PSI[1]) show shift signal for *all* 8 tasks uniformly (MMD $p < 0.002$; C2ST >99.9% accuracy; PSI > 0.002), but cannot distinguish catastrophic from robust outcomes ($\rho \leq 0.19$, all $p > 0.6$). Among evaluated pre-deployment metrics, SHAP concentration best captures the model-specific dependence structure linked to failure severity.

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees; Shafer and Vovk [2008] provide an accessible tutorial. Recent work extends to classification [Romano et al., 2020, Cauchois et al., 2021] and regression [Romano et al., 2019, Lei et al., 2018]. Angelopoulos and Bates [2023] provide a comprehensive survey. Ding et al. [2023] show that class-conditional coverage guarantees become harder to achieve as class count grows, and propose clustered conformal prediction as a remedy—a confound we explicitly examine via partial correlation analysis (Section 5.4).

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free uncertainty quantification for classification under label shift.

[1] Population Stability Index, a monitoring metric standard in credit scoring practice [Siddiqi, 2006].

Barber et al. [2023] provide a comprehensive treatment beyond exchangeability, bounding coverage loss by the total variation distance between test and calibration score distributions. Kasa and Taylor [2023] empirically characterize how CP degrades across many vision architectures and shift types, showing substantial degradation is common—our work addresses the complementary question of which tabular models will fail *before* test data is observed. We complement these theoretical characterizations with an empirical diagnostic that is associated with this distributional divergence—and hence with which tasks fail—before deployment.

Adaptive Methods. Gibbs and Candès [2021, 2024] develop Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series, Feldman et al. [2023] for online risk control, and Bhatnagar et al. [2023] for strongly adaptive online learning. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Kasa et al. [2025] propose entropy-based methods (ECP/EACP) that adapt prediction sets using unlabeled test data at deployment time—a related but distinct goal from ours, which requires no test observations: SHAP concentration is a structural pre-deployment diagnostic computed solely on validation data. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show domain generalization methods often fail to improve over ERM under realistic distribution shifts. Gardner et al. [2023] benchmark tabular distribution shift across 15 tasks—the closest tabular-specific benchmark; our SALT dataset differs in providing a naturalistic COVID-19 temporal split with documented business-disruption context rather than curated benchmark partitions. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Kernel-based tests (MMD [Gretton et al., 2012]) and classifier two-sample tests (C2ST [Lopez-Paz and Oquab, 2017]) detect distributional differences but do not predict the *severity* of downstream failure. We show empirically that MMD and C2ST detect shift for all tasks uniformly yet cannot distinguish catastrophic from robust conformal outcomes (Section 6.3). Our contribution is a *pre-deployment* diagnostic using feature importance structure that predicts failure severity, not merely shift presence.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017, Lundberg et al., 2020] to model reliability assessment, extending the use of feature attribution from post-hoc model analysis to prospective failure diagnosis. Prior work uses SHAP reactively—monitoring attribution changes post-deployment to detect concept drift [Adebayo et al., 2018]. Our use is prospective: SHAP concentration is a static structural property computed pre-deployment on validation data, predicting which models will fail rather than detecting failures that have already occurred.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocaTion) dataset from RelBench [Robinson et al., 2024], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded. External validation uses 9 additional datasets spanning 9 non-supply-chain benchmarks. The primary endpoint includes only $K \geq 4$ class tasks: near-binary tasks ($K \leq 3$) are excluded because binary APS prediction sets ($\{0\}$, $\{1\}$, $\{0, 1\}$) have a structural ceiling that blocks the concentration mechanism (Contribution 5); this criterion follows from APS mechanics, not from the data. Stack Overflow ($K = 3$) falls under this exclusion. Applying it yields 8 external multiclass datasets; together with the 8 multiclass SALT tasks this gives the primary endpoint of $n = 16$ multiclass tasks across 9 domains (SALT counts as one domain).

- **Train**: Before February 2020 (pre-COVID)
- **Validation**: February–July 2020 (COVID onset)
- **Test**: After July 2020 (COVID peak)

Data separation protocol. SHAP concentration and the 40% threshold are computed exclusively on validation-time data. The test set is used only for final evaluation and never informs diagnostic thresholds. For threshold evaluation we use at-risk labels defined by observed test coverage drops (drop $>$15 pp); the threshold sensitivity analysis in Appendix D is therefore post-hoc validation, not prospective assessment.

3.2 CONFORMAL PREDICTION SETUP

We use Adaptive Prediction Sets (APS) [Romano et al., 2020] with $\alpha = 0.1$ (90% target coverage). We use the non-randomized variant (without the uniform tie-breaking variable U), which provides conservative rather than exact coverage. This choice does not affect the diagnostic analysis, as the same scoring rule is applied to both calibration and test sets.

1. Train LightGBM classifier with fixed hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

We train 50 independent models with seeds 42–91. For each seed s: train model M_s, split validation 50/50 into calibration $\mathcal{D}^s_{\text{cal}}$ and evaluation $\mathcal{D}^s_{\text{eval}}$, calibrate CP$_s$ on $\mathcal{D}^s_{\text{cal}}$, evaluate on both $\mathcal{D}^s_{\text{eval}}$ and test. We report mean, 95% confidence intervals (t-distribution), and paired Wilcoxon tests (seed-level). This is *not* ensemble prediction—each seed is an independent experimental trial providing paired comparisons.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(x_j) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \tag{1}$$

where A_{train} and A_{test} are the sets of unique values for feature x_j in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^{p} \phi_j} \tag{2}$$

where ϕ_j is the mean absolute SHAP value of the j-th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 2), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). Among the diagnostics we evaluate, SHAP concentration is available before deployment and shows the strongest severity association.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? We compared top-1 concentration against alternatives: top-2, top-3, Herfindahl-Hirschman Index (HHI), and feature importance entropy. Top-1 concentration is the only variant among these five that reaches nominal significance with coverage drop ($\rho = 0.833$, $p = 0.010$); all alternatives are non-significant ($p > 0.10$). Under Holm–Bonferroni correction for five metrics, the adjusted p-value for top-1 is 0.050 (boundary; Appendix G); the $n = 16$ cross-domain result ($\rho = 0.853$, $p < 0.001$) was not subject to this selection correction and provides independent replication. The discriminating power of top-1 arises because sales-office has high top-2/3 concentration (77–85%) but near-zero coverage drop—the diagnostic signal is specifically whether a *single* dominant feature drives predictions, not overall feature spread.

4 THEORY: WHY CONCENTRATION RELATES TO FAILURE

We establish a mechanistic account for why SHAP concentration is associated with conformal failure under feature shift. The following theorem shows that when a model concentrates dependence on a single feature that shifts out of distribution, APS score and coverage bounds worsen monotonically with concentration parameter C, under explicit assumptions.

Notation. For a predicted distribution $\hat{p}(y \mid x) \in \Delta^{K-1}$, the APS conformity score $s(x, y^*)$ sums predicted probabilities from the most likely class down to and including y^*. Equivalently:

$$s(x, y^*) \;=\; 1 \;-\; \sum_{\substack{y \neq y^*: \\ \hat{p}(y|x) < \hat{p}(y^*|x)}} \hat{p}(y \mid x), \tag{3}$$

since s includes all mass at or above the rank of y^*, leaving out only classes ranked strictly below. (The bound holds regardless of randomized tie-breaking conventions in APS implementations, since $\mathcal{B}$ uses strict inequality.)

Theorem 1 (Score Inflation Under Concentrated Feature Shift (Idealized Additive Model))**.** *Consider a K-class classifier ($K \geq 3$) with APS conformal predictor calibrated at level α with quantile $\hat{q}_\alpha$. Let $C \in [0, 1]$ denote SHAP concentration on feature x_1. Assume:*

(A1) ***Additive decomposition.*** $\hat{p}(y \mid x) = C \cdot g(y \mid x_1) + (1 - C) \cdot h(y \mid x_{\setminus 1})$, *where* $g(\cdot \mid x_1), h(\cdot \mid x_{\setminus 1}) \in \Delta^{K-1}$.[2]

[2] Assumption (A1) posits additivity in *probability* space; TreeExplainer computes SHAP in *log-odds* space, so (A1) is an idealization. The theorem establishes directional intuition—monotone score inflation under concentrated dependence—rather than a formal guarantee for the measured SHAP-derived C. The empirical proxy is validated by $\rho = 0.853$ and conservative-bound verification (Appendix H).

(A2) ***Concentrated misclassification.*** *Under shift,* $g(y^* \mid x_1^{\text{test}}) \le \varepsilon$ *for some* $\varepsilon \in [0, 1/K)$.

(A3) ***Residual exchangeability.*** $(h(\cdot \mid x_{\setminus 1}^{\text{test}}), y^*_{\text{test}}) \overset{d}{=} (h(\cdot \mid x_{\setminus 1}^{\text{cal}}), y^*_{\text{cal}})$.

Then:

(i) ***Pointwise score bound.*** *For each test instance* (x^{test}, y^*)*:*

$$s(x^{\text{test}}, y^*) \;\ge\; 1-(K-1)\big[C\varepsilon+(1-C)\,h(y^*\,|\,x_{\setminus 1}^{\text{test}})\big]. \quad (4)$$

(ii) ***Expected score inflation.*** *Let* $\bar{h} := \mathbb{E}[h(y^*_{\text{cal}} \mid x_{\setminus 1}^{\text{cal}})]$ *denote the expected true-class probability under the residual model. Then:*

$$\mathbb{E}[s_{\text{test}}] \;\ge\; C(1-(K-1)\varepsilon) \;+\; (1-C)(1-(K-1)\bar{h}). \quad (5)$$

(iii) ***Monotone vulnerability.*** *The bound in* (5) *is strictly increasing in* C *whenever* $\varepsilon < \bar{h}$. *A sufficient condition is* $\bar{h} \ge 1/K$ *(the residual model is at least uniform on the true class); together with* (A2) *this implies* $\varepsilon < \bar{h}$.

(iv) ***Coverage degradation bound.*** *Coverage satisfies:*

$$\mathbb{P}(y^* \in \mathcal{C}(x^{\text{test}})) \;\le\; \mathbb{P}(h(y^*_{\text{cal}}) \ge T(C))\,, \quad (6)$$

where $T(C) := \big[(1-\hat{q}_\alpha)/(K-1) - C\varepsilon\big]/(1-C)$ *is strictly increasing in* C *when* $\varepsilon < (1-\hat{q}_\alpha)/(K-1)$ *(satisfied at* $\varepsilon = 0$*). Hence the coverage upper bound is non-increasing in* C.

Proof sketch. Part (i) upper-bounds omitted APS mass by $(K-1)\hat{p}(y^*)$ and substitutes assumptions (A1)–(A2). Part (ii) takes expectations and uses residual exchangeability (A3). Part (iii) differentiates the lower bound in (5) and gives monotonicity when $\varepsilon < \bar{h}$. Part (iv) rearranges (4) to obtain (6) via threshold $T(C)$ and shows $T'(C) > 0$ under the stated condition. $\square$

The full derivation and conservative bound verification are in Appendix H. Empirically, catastrophic tasks show strong rightward conformity-score shifts (KS = 0.68–0.96), while concentration and shift magnitude are weakly related across SALT ($\rho = 0.29$, $p = 0.49$). This supports interpreting concentration as a model-vulnerability proxy (dependence structure) rather than a shift-size metric.

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing identical COVID-19 temporal shift, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 1). All coverage drops are statistically significant (paired Wilcoxon $p \le 0.005$, 95% CIs from the t-distribution), yet range from 0.1% to 77.1%.

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 2).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Across all 8 SALT multiclass tasks, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 3, Figure 2). Bootstrap 95% CI: $[0.29, 1.00]$ (10,000 resamples). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 1 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion (4.5× increase). Robust tasks: importance redistributes across features with ranking reshuffle (10–20× increases distributed). This matches the prediction of Theorem 1.

False positives. Two tasks with concentration >40% remain robust: sales-office (secondary feature SALESORG has Jaccard = 0.61) and item-shippoint (high model variance). These motivate the protective-factor check in our framework (Section 7).

Task independence. Intraclass correlation analysis confirms task identity explains the majority of seed-level variance in coverage drops (ICC = 0.675, 95% CI $[0.47, 0.90]$; effective $n_{\text{eff}} = 11.7$). Per-task ICC(val,test) is negative for all 8 tasks, indicating paired tests are conservative. Details in Appendix G.

Statistical power. At $n = 8$, power to detect $\rho = 0.833$ is ≈ 0.76 (below the 0.80 convention); the within-SALT result is therefore *exploratory*. The $n = 16$ cross-domain result ($\rho = 0.853$, $p < 0.001$; power > 0.99) is the confirmatory endpoint.

Cross-domain extension. Extending to 16 multiclass tasks across 9 domains—including 8 external datasets spanning diverse types of natural distribution shift (Section 6.5)—yields $\rho = 0.853$, $p < 0.001$ (Kendall $\tau = 0.667$; bootstrap 95%

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t-distribution. p-values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	Cl	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	462	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	135	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	70	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB*†
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV $>$ 50%), classified by median; †meets At-risk criterion (drop >15 pp) under the decision framework (Table 8).

Table 2: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val $-$ mean test coverage (50 seeds).

Table 3: Stratified Correlation Analysis.

Group	n	ρ	Kendall τ	$p(\rho)$	Boot. 95% CI
Multiclass (SALT)	8	0.833	0.714	0.010	$[0.29, 1.00]$
Multiclass (4 dom.)	11	0.909†	0.782	<0.001	$[0.61, 1.00]$
Multiclass (8 dom.)	15	0.882‡	0.714	<0.001	$[0.60, 0.97]$
Multiclass (9 dom.)	**16**	**0.853**	**0.667**	**<0.001**	$[\mathbf{0.50}, \mathbf{0.96}]$
Combined (11 dom.)	19	0.814	0.626	<0.001	—

LOO stability (SALT): $\rho \in [0.75, 0.96]$; 6/8 jackknife significant. Multiclass (9 domains) is the primary result, excluding binary ceiling-effect tasks. Under the at-risk criterion (drop >15 pp), threshold performance at 40% is Precision $= 0.83$, Recall $= 0.83$ (Appendix D). †Single-seed external values; multi-seed-consistent value at $n = 11$ is $\rho = 0.818$, $p = 0.002$ (see Appendix G). $^{\ddagger}n = 15$ is an intermediate analysis (7 external multiclass domains); $n = 16$ adds KDDCup99 as the 8th external domain to complete the 9-domain primary endpoint. Stack Overflow (3 classes, near-binary ceiling effect) is excluded from all multiclass endpoints.

CI $[0.50, 0.96]$). We note that the 40% threshold was derived from the SALT subset ($n = 8$); the $n = 16$ correlation therefore includes the in-sample SALT tasks ($\rho = 0.833$ within SALT alone) alongside the held-out external 8 datasets on which the threshold was not tuned. The external datasets provide the cleanest out-of-sample validation: Covertype is correctly flagged as catastrophic (81.8 pp drop, 10/10 seeds), 6 deterministic low-concentration datasets remain robust (10/10 seeds each), and KDDCup99 is the principal intermediate-regime false negative.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 4).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds (seed 42). Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as confound. Within SALT, $\log(\text{num\_classes})$ alone correlates with coverage drop ($\rho = 0.743$, $p = 0.035$), and both partial correlations are non-significant at $n = 8$ ($\rho_{\text{partial}} = 0.629$, $p = 0.131$ for concentration; 0.334, $p = 0.464$ for cardinality; Table 4). This non-significance is expected at $n = 8$—within-SALT the two predictors are correlated and power is insufficient for causal separation. Cross-domain evidence provides complementary support: Covertype (7 classes, $C = 49.8\%$) fails catastrophically, while higher-cardinality low-concentration datasets such as Avila (12 classes, $C = 20.5\%$), PAMAP2 (12, 19.2%), and Pendigits (10, 14.5%) remain robust—showing that high class count alone does not predict failure. KDDCup99 (11, 21.1%) is a seed-dependent intermediate case rather than a clean robust counterexample. At $n = 16$ (8 SALT + 8 external multiclass tasks), the partial Spearman controlling for $\log K$ yields $\rho_{\text{partial}}(\text{conc.}) = 0.771$ ($p = 0.0008$) while $\rho_{\text{partial}}(\log K) = -0.010$ ($p = 0.97$), providing strong cross-domain evidence that concentration drives the severity association independently of class count.

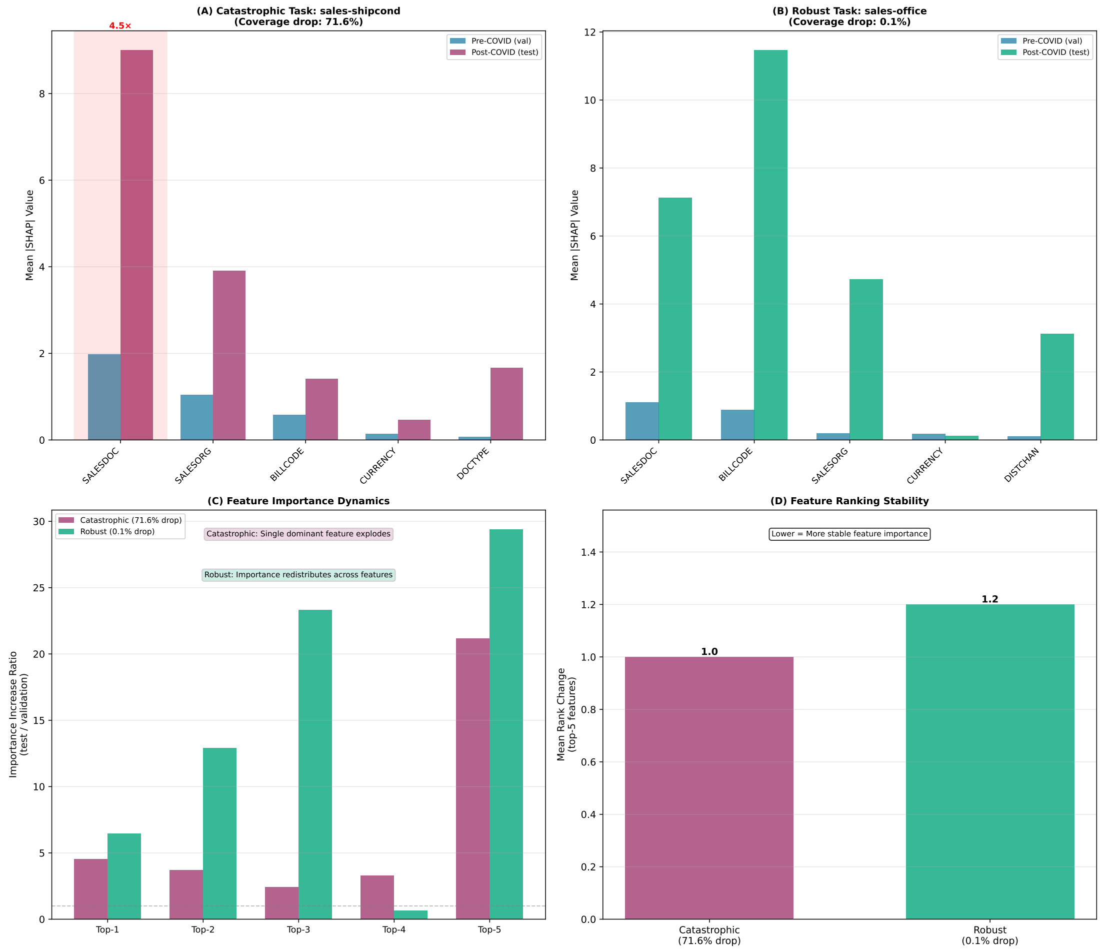

Figure 1: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes 4.5× while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite ∼0% Jaccard similarity, coverage drops range from negligible to catastrophic.

Table 4: Pre-Deployment Diagnostic Comparison ($n = 8$). Spearman ρ with 50-seed mean coverage drop. ρ_{partial}: partial Spearman controlling for the other variable in the SHAP–class-count pair.

Diagnostic	ρ	p	ρ_{partial}	p_{partial}
SHAP concentration	**0.833**	**0.010**	0.629	0.131
log(num_classes)	0.743	0.035	0.334	0.464
Native FI concentration	0.667	0.071	—	—
Ensemble disagreement	0.452	0.260	—	—

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

Expected Calibration Error change: $\Delta\text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6 EXTENDED EXPERIMENTS

6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

ACI [Gibbs and Candès, 2021] improves nominal coverage on severe tasks but often with large prediction-set inflation. Coverage gains are +47.5 pp (s-shipcond), +71.8 pp (s-group), and +60.0 pp (s-payterms); usable-set rates are 30%, 82%, and 40%, respectively. Thus, only s-shipcond remains clearly deployable after adaptation. On robust tasks, ACI gains are small or non-significant (+1.2 to +10.1 pp),

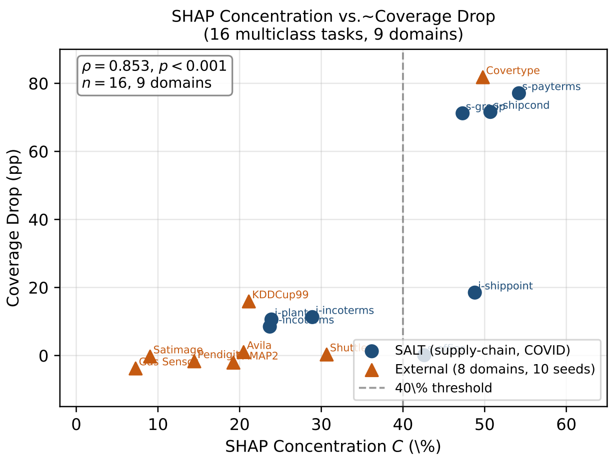

Figure 2: **SHAP Concentration Correlates with Coverage Degradation (Primary Result).** Spearman $\rho = 0.853$, $p < 0.001$ across all 16 multiclass tasks in 9 domains (dark circles: 8 SALT supply-chain tasks; orange triangles: 8 external domains, 10-seed means). The 40% threshold (dashed) is derived from validation data only. Within SALT alone: $\rho = 0.833$, $p = 0.010$ ($n = 8$).

and sales-office shows slight degradation from near-ceiling coverage. The practical implication is that online adaptation can recover coverage, but often by sacrificing informativeness in the highest-risk regimes.

6.2 ALTERNATIVE CONFORMAL SCORING

RAPS [Angelopoulos et al., 2021] shows mechanism-dependent benefit. For high-cardinality tasks, it can substantially reduce drops (s-group: $73.5 \pm 33.9\% \to 10.4 \pm 1.3\%$; s-payterms: $79.9 \pm 29.4\% \to 35.1 \pm 28.6\%$). For the most concentrated task (s-shipcond, $C = 50.7\%$), RAPS does not help ($60.4 \pm 31.8\% \to 67.8 \pm 25.4\%$). This pattern matches Section 4: regularization can control class-accumulation effects, but not failures driven by concentrated single-feature dependence. Full task-wise results are in Appendix J.

6.3 COMPARISON WITH STANDARD SHIFT DETECTION

We evaluate MMD, C2ST, and PSI as pre-deployment alternatives. All detect shift on all 8 SALT tasks (MMD $p < 0.002$, C2ST $> 99.9\%$ accuracy, PSI > 0.002), but none predict severity: correlations with coverage drop are $\rho = -0.048$ (MMD), $\rho = 0.191$ (C2ST), and $\rho = 0.071$ (PSI), all non-significant. This reinforces the distinction between detecting distribution change and predicting model-specific failure amplification.

6.4 RETRAINING ANALYSIS

Retraining helps selectively. Quarterly retraining improves sales-shipcond by +18.9 pp ($p = 0.04$, unadjusted; Holm-corrected over the 3 severe tasks evaluated: $p = 0.12$), partially improves sales-payterms (+15.5 pp), and is ineffective for sales-group (462 classes; $0.0\% \to 2.2\%$). Robust tasks remain near-ceiling without intervention. Thus, retraining is beneficial for some vulnerable moderate-cardinality tasks but does not resolve the highest-complexity failures.

6.5 CROSS-DOMAIN AND EXTERNAL VALIDATION

External validation extends to 9 additional datasets across 9 non-supply-chain benchmarks [Dua and Graff, 2017] with the same LightGBM+APS pipeline and no parameter tuning. These datasets span diverse shift mechanisms: geographic (Covertype), temporal batch drift (Gas Sensor Array), attack-type distribution shift (KDDCup99), semantic/category shift (Stack Overflow), and pre-defined benchmark splits whose shift type is inherent in the UCI partition (Shuttle, Avila, PAMAP2, Pendigits, Satimage); see Appendix A.6 for full protocols. On the multiclass primary set ($n = 16$, 9 domains), concentration remains strongly associated with failure severity ($\rho = 0.853$, $p < 0.001$; Kendall $\tau = 0.667$). Covertype is the key external catastrophic case ($C = 49.8\%$, 81.8 pp drop, 10/10 seeds), while low-concentration datasets are mostly robust (6 deterministic robust domains and 1 near-deterministic domain). KDDCup99 is the main intermediate-regime exception (mean drop $= 15.9 \pm 21.4$ pp despite $C = 21.1\%$), motivating the conservative uncertainty-band policy in Section 7. Binary and near-binary ceiling-effect tasks exhibit the expected ceiling effect and are excluded from the multiclass primary endpoint.

6.6 PLACEBO TEST

Using a pre-COVID temporal placebo (train 2018, validate 2019-H1, test 2019-H2), vulnerable tasks show far smaller degradation than under the COVID split (6–143$\times$ lower; Appendix B). This supports the claim that the diagnostic is most informative under severe event-driven shift rather than routine temporal drift.

7 DECISION FRAMEWORK

The following exploratory protocol, empirically motivated by our analysis, can guide practitioners deploying conformal prediction in non-stationary settings:

1. **Compute SHAP concentration** $C = \phi_1 / \sum_j \phi_j$ on validation data (Eq. 2)

2. **If** $C \leq 40\%$: lower observed risk in this study (distributed importance may protect against shift)

3. **If** $C > 40\%$: Check for protective factors—secondary features with train-validation Jaccard > 0.5 *and* importance $> 15\%$ (validation overlap serves as a pre-deployment proxy for deployment-time feature stability)

   - Protective factors exist → **robust**
   - Otherwise → **vulnerable**: consider quarterly retraining (effect suggestive; Holm-corrected $p = 0.12$ over the 3 severe tasks evaluated)

4. **Safety override**: if C is in the 35–45% ambiguity band or the multi-seed 95% CI overlaps 40%, treat the task as **uncertain risk** and enable intensified monitoring rather than assuming robustness

5. **For lower-risk tasks**: defer routine retraining (saves computational cost)

6. **Monitor**: Track empirical coverage as early warning

Seed stability protocol. KDDCup99's multi-seed analysis ($C = 21.1 \pm 7.5\%$) reveals that single-seed SHAP concentration can be unstable. We recommend computing C over $k \geq 5$ seeds and using the mean. Tasks in the ambiguous zone (35–45%) should receive additional scrutiny: if the upper 95% CI exceeds 40%, treat as potentially vulnerable. This safety override is designed to reduce KDDCup99-like false negatives in intermediate regimes. For SALT tasks, concentration is stable across 50 seeds (within-task CV $<$ 5%); instability arises primarily when the top-feature identity varies across seeds.

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 8. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. The 40% threshold applied without tuning to 9 external domains yields 7/9 deterministic or near-deterministic outcomes (6 at 10/10, Shuttle at 9/10), with Covertype correctly flagged as catastrophic and Gas Sensor correctly unflagged as robust. KDDCup99 is the principal intermediate-regime false negative, while Stack Overflow (3 classes) exhibits the near-binary ceiling effect. Across the $n = 16$ multiclass set: $\rho = 0.853$, $p < 0.001$ (Section 6.5). The sensitivity analysis in Appendix D reports threshold performance across 30–50%; given sparse catastrophic external cases, this threshold should be treated as provisional.

8 DISCUSSION

Catastrophic tasks become *confidently wrong*: prediction sets can shrink while coverage collapses, consistent with concentrated dependence on OOD features. This explains why generic shift detectors are insufficient for deployment triage: they detect shift everywhere, while concentration predicts which models amplify shift into failure.

Limitations. This is associative evidence, not causal identification. The 40% threshold is exploratory (derived on $n = 8$ multiclass SALT tasks) and should be treated as provisional despite external support. Evidence is model-specific (strongest for boosting models), and external catastrophic support is sparse (dominated by Covertype), so prospective deployment validation remains necessary.

9 CONCLUSION

SHAP concentration provides a pre-deployment, model-specific signal for conformal vulnerability under shift. Across 16 multiclass tasks in 9 domains, concentration tracks failure severity ($\rho = 0.853$, $p < 0.001$), while standard shift detectors do not separate catastrophic and robust outcomes ($\rho \leq 0.19$). The theorem formalizes monotone score inflation under concentrated dependence, and secondary experiments show mitigation is mechanism-dependent (RAPS helps many-class accumulation failures; retraining/ACI can recover coverage but may hurt utility). The proposed deployment policy is therefore conservative: use concentration with uncertainty-band overrides (35–45%), protective-factor checks, and monitoring, rather than treating any fixed threshold as universal.

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

Anastasios N Angelopoulos, Stephen Bates, Jitendra Malik, and Michael I Jordan. Uncertainty sets for image classifiers using conformal prediction. In *International Conference on Learning Representations*, 2021.

Anastasios N Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations*, 2024.

Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.

Aadyot Bhatnagar, Huan Wang, Caiming Xiong, and Yu Bai. Improved online conformal prediction via strongly adaptive online learning. In *International Conference on Machine Learning*, pages 2337–2363. PMLR, 2023.

Maxime Cauchois, Suyash Gupta, and John C Duchi. Knowing what you know: Valid and validated confidence sets in multiclass and multilabel prediction. *Journal of Machine Learning Research*, 22(81):1–42, 2021.

Thomas Ding, Anastasios N Angelopoulos, Stephen Bates, Michael I Jordan, and Ryan J Tibshirani. Class-conditional conformal prediction with many classes. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Dheeru Dua and Casey Graff. UCI machine learning repository. University of California, Irvine, School of Information and Computer Sciences, 2017. URL `https://archive.ics.uci.edu`.

Shai Feldman, Liran Ringel, Stephen Bates, and Yaniv Romano. Achieving risk control in online learning settings. *Transactions on Machine Learning Research*, 2023.

Josh Gardner, Zoran Popovic, and Ludwig Schmidt. TableShift: Benchmarking the robustness of tabular learning methods. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.

Saurabh Garg, Sivaraman Balakrishnan, Zachary C Lipton, Behnam Neyshabur, and Hanie Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. In *International Conference on Learning Representations*, 2022.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, pages 1660–1672, 2021.

Isaac Gibbs and Emmanuel Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36, 2024.

Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13: 723–773, 2012.

Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021.

Siddharth Ravi Kasa and Michael E Taylor. Empirically validating conformal prediction on modern vision architectures under distribution shift and long-tailed data. *arXiv preprint arXiv:2307.01088*, 2023.

Siddharth Ravi Kasa, Arwa Alanqary, Fredy Diaz, and Promit Ghosal. Adapting prediction sets to distribution shifts without labels. *arXiv preprint arXiv:2406.01416*, 2025.

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. In *International Conference on Machine Learning*, pages 5637–5664. PMLR, 2021.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.

David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*, 2017.

Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Scott M Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

Andrey Malinin, Neil Band, German Chesnokov, Yarin Gal, Mark JF Gales, Alexey Noskov, Andrey Ploskonosov, Liudmila Prokhorenkova, Ivan Provilkov, Vatsal Raina, et al. Shifts: A dataset of real distributional shift across multiple large-scale tasks. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.

Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free uncertainty quantification for classification under label shift. In Cassio de Campos and Marloes Maathuis, editors, *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 844–853. PMLR, 2021. URL `https://proceedings.mlr.press/v161/podkopaev21a.html`.

Joshua Robinson, Rishabh Ranjan, Weihua Hu, Kexin Huang, Jiaqi Han, Alejandro Dobles, Matthias Fey, Jan Eric Lenssen, Yiwen Yuan, Zecheng Zhang, Xinwei He, and Jure Leskovec. Relbench: A benchmark for deep learning on relational databases. In *Advances in Neural Information Processing Systems Track on Datasets and Benchmarks*, 2024.

Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Yaniv Romano, Matteo Sesia, and Emmanuel Candès. Classification with valid and adaptive coverage. In *Advances in Neural Information Processing Systems*, volume 33, pages 3581–3591, 2020.

Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9: 371–421, 2008.

Naeem Siddiqi. *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. Wiley, 2006.

Ryan J Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Margaux Zaffran, Olivier Féron, Yannig Goude, Julie Josse, and Aymeric Dieuleveut. Adaptive conformal predictions for time series. In *International Conference on Machine Learning*, pages 25834–25866, 2022.

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

A REPRODUCIBILITY DETAILS

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with fixed LightGBM settings: objective=multiclass, boosting=gbdt, num_leaves=31, learning_rate=0.05, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5, num_boost_round=500, early_stopping_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage: $(1 - \alpha) = 0.9$ (90%). Calibration: random 50% split of validation set. Quantile: $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$.

Calibration and test set sizes. Sales-level tasks (s-shipcond, s-payterms, s-group, s-incoterms, s-office): $n_{\text{val}} = 71{,}474$, $n_{\text{cal}} = 35{,}737$, $n_{\text{test}} = 88{,}942$. Item-level tasks (i-plant, i-shippoint, i-incoterms): $n_{\text{val}} = 293{,}896$, $n_{\text{cal}} = 146{,}948$, $n_{\text{test}} = 402{,}855$. At these calibration set sizes, the finite-sample correction is negligible: $q - (1 - \alpha) < 3 \times 10^{-5}$ for all tasks, so finite-sample effects do not contribute to the observed coverage gaps.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM's gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: $\sim$3–4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

A.6 EXTERNAL DATASET PROTOCOLS

Table 5 documents the train/test split, shift mechanism, and dataset characteristics for all 9 external validation datasets. All use the same LightGBM+APS pipeline as SALT (Appendix A), with 10 seeds and no hyperparameter tuning.

Table 5: External Dataset Protocols. Cal = calibration set (50% of train); n_{tr}, n_{te} = approximate train/test sizes used; Seeds = 10 for all external datasets.

Dataset	Shift mechanism
Covertype	Geographic: wilderness area 1 (train) vs. areas 2–4 (test)
Gas Sensor	Temporal batch drift: batches 1–6 (train), 7–8 (cal), 9–10 (test)
KDDCup99	Attack-type distribution shift; KDD Cup 1999 competition 10% subset (`kddcup.data_10_percent`), official train/test files
Shuttle	UCI Statlog pre-defined split
Avila Bible	UCI pre-defined split (copyist identification)
PAMAP2	UCI pre-defined split (physical activity, sensor data)
Pendigits	UCI pre-defined split (pen-based digit recognition)
Satimage	UCI pre-defined split (satellite image classification)
Stack Overflow	Category shift (3-class subset); excluded from $n = 16$ primary endpoint (near-binary ceiling effect)

For Covertype, train uses wilderness area 1 only; test uses all remaining areas. For Gas Sensor, each batch contains ∼1,000 samples across 6 gas types; temporal sensor drift is the documented shift mechanism. For KDDCup99, classes are the 11 attack categories present in the 10% subset; the standard KDD Cup 1999 train/test split is used. For UCI pre-defined split datasets (Shuttle, Avila, PAMAP2, Pendigits, Satimage), the official UCI train and test partitions are used directly; the shift type is inherent in the benchmark partition and not separately documented in the UCI repository. SHAP concentration and conformal calibration use the UCI train partition only.

B PLACEBO TEST DETAILS

Table 6: Placebo Test: Pre-COVID vs COVID Degradation (50-seed mean drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 1 exhibit extreme model variance (CV $>$ 50%):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.
- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

D THRESHOLD SENSITIVITY ANALYSIS

Table 7 shows threshold performance across $n = 16$ multiclass tasks. Note that the 40% threshold is derived from the SALT concentration gap (in-sample); the 8 external datasets are fully out-of-sample. On the external subset alone at 40%: Precision $=$ 1.00, Recall $=$ 0.50 (1 TP: Covertype; 1 FN: KDDCup99; 6 TN). Low external recall reflects sparse catastrophic cases; the primary claim rests on the $n = 16$ correlation ($\rho = 0.853$) rather than threshold classification. Ground-truth labels use the at-risk rule (coverage drop $>$15 pp).

Table 7: Threshold Sensitivity for SHAP Concentration across 16 multiclass tasks (8 SALT + 8 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop $>15\%$.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.63	0.83	0.71	5	3	1
30%	0.71	0.83	0.77	5	2	1
35%	0.83	0.83	0.83	5	1	1
40%	**0.83**	**0.83**	**0.83**	**5**	**1**	**1**
45%	1.00	0.83	0.91	5	0	1
50%	1.00	0.33	0.50	2	0	4

At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard $= 0.61$), and the single FN is KDDCup99 (intermediate regime; $C = 21.1\%$, drop $= 15.9$ pp). At 30%, the two FPs are s-office and Shuttle ($C = 30.7\%$); at 25%, i-incoterms ($C = 28.9\%$) is additionally flagged. Applying the protective-factor check at 40% removes s-office (precision $= 1.00$) but leaves the KDDCup99 FN (recall $= 0.83$).

E DECISION FRAMEWORK VALIDATION

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Counter-intuitively, all three catastrophic tasks (coverage drop $> 70\%$) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). In this SALT comparison, SHAP concentration is the only evaluated pre-deployment metric that flags all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. For coverage drops, ICC $= 0.675$ (95% CI: $[0.47, 0.90]$; $F(7, 392) = 104.6$), confirming that task identity explains the majority of seed-level variance. The design effect is 34.1, yielding effective $n_{\text{eff}} = 400/34.1 = 11.7$—exceeding the 8 task-level observations. For test coverages, ICC $= 0.720$ (95% CI: $[0.52, 0.92]$). These values confirm that the $n = 8$ tasks constitute meaningfully independent observations for the Spearman correlation analysis.

Per-task ICC(val,test). For each task, we compute ICC between the 50 validation coverages and 50 test coverages (paired by seed). All 8 per-task ICCs are negative (-0.80 to -0.10), indicating that seeds producing above-average validation coverage do *not* produce above-average test coverage. This means paired Wilcoxon tests are conservative: the effective sample size for within-task paired tests *exceeds* 50, so the reported p-values are, if anything, too large.

Multiple comparisons. We tested 5 concentration metrics (top-1, top-2, top-3, HHI, entropy) against coverage drop. Under Holm–Bonferroni correction for this 5-test family, the adjusted p-value for top-1 concentration is $5 \times 0.010 = 0.050$, at the conventional significance boundary. For the primary correlation subsets ($n = 8$, $n = 11$, $n = 16$), all adjusted p-values remain significant (Holm-corrected $p \leq 0.021$). We note that top-1 was the a priori natural choice (Section 3.6), not one of 5 equally plausible alternatives; the correction is therefore conservative.

Multi-seed aggregation. The primary $n = 16$ correlation ($\rho = 0.853$) uses 10-seed means for all external datasets and 50-seed means for SALT tasks. The early $n = 11$ result ($\rho = 0.909$) used single-seed external values; switching to multi-seed means at $n = 11$ yields $\rho = 0.818$, $p = 0.002$, with the reduction driven by KDDCup99 whose multi-seed mean drop (15.9 pp) exceeds its single-seed value (-0.8 pp). This intermediate-concentration regime ($C = 21.1 \pm 7.5\%$) exhibits high outcome variance characteristic of the boundary between robust and vulnerable. Extending to $n = 16$ with consistent

Table 8: Concentration Check Applied to All Multiclass Tasks (Actual = At-risk if drop >15 pp).

Task	Conc. (%)	Step 2	Actual (>15 pp)	Note
SALT (in-sample)				
s-payterms	54.2	VULN	At-risk	—
s-shipcond	50.7	VULN	At-risk	—
i-shippoint	48.8	VULN	At-risk*	High variance†
s-group	47.3	VULN	At-risk*	—
s-office	42.6	VULN	ROB	Protective‡
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—
External (held-out)				
Covertype	49.8	VULN	At-risk	10/10 seeds
Shuttle	30.7	ROB	ROB	9/10 seeds
Avila Bible	20.5	ROB	ROB	10/10 seeds
PAMAP2	19.2	ROB	ROB	10/10 seeds
KDDCup99	21.1	ROB	At-risk§	Seed-dependent (FN)
Pendigits	14.5	ROB	ROB	10/10 seeds
Satimage	9.0	ROB	ROB	10/10 seeds
Gas Sensor	7.3	ROB	ROB	10/10 seeds
Stack Overflow	48.9	VULN	ROB	Near-binary ceiling¶

†Median drop 1.2%; most seeds maintain coverage. ‡SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%. §Concentration below threshold ($C = 21.1 \pm 7.5\%$), but mean drop exceeds the at-risk cutoff (15.9 pp); outcome seed-dependent (range -0.8 to 73.5 pp). In-sample F1 $= 0.89$ (1 FP: s-office). External: Covertype deterministic catastrophic (10/10); five low-concentration datasets deterministic robust (10/10); Shuttle near-deterministic (9/10); KDDCup99 is the intermediate-regime false negative. ¶Stack Overflow ($K = 3$, $C = 48.9\%$): high concentration predicts VULN, but near-binary class structure creates a ceiling effect (only 3 classes; APS prediction sets trivially remain small). Actual coverage is robust (-7.0 pp). Excluded from $n = 16$ multiclass primary endpoint; confirms diagnostic applies to multiclass ($K > 3$) settings only.

Table 9: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-seed ACI experiments). SHAP concentration is computed on validation data only. ΔEntropy and ΔECE require test-time observations.

Task	Cat	SHAP Conc.†	ΔEntropy	ΔECE
Severe tasks				
s-shipcond	SEV	50.7%	−0.746	+0.408
s-group	SEV	47.3%	−1.274	+0.176
s-payterms	SEV	54.2%	−1.979	+0.629
Robust tasks				
i-plant	ROB	23.9%	+0.181	+0.1314
i-shippoint	ROB	48.8%	+0.469	+0.0727
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

†Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

Table 10: Per-Task ICC Between Validation and Test Coverages (50 Seeds).

Task	ICC(val,test)	Design Effect	Eff. n
s-payterms	−0.80	0.20	250
s-shipcond	−0.77	0.23	220
s-group	−0.72	0.28	177
i-plant	−0.42	0.58	86
i-incoterms	−0.37	0.63	80
s-incoterms	−0.35	0.65	77
s-office	−0.15	0.85	59
i-shippoint	−0.10	0.90	55

multi-seed means yields $\rho = 0.853$ ($p < 0.001$). Including Stack Overflow (3 classes, near-binary ceiling) to make $n = 17$ weakens correlation to $\rho = 0.654$.

H THEORY DETAILS AND SCORE CDFS

Proof details. Starting from (3), define $L := \sum_{y \in \mathcal{B}} \hat{p}(y)$ for classes below the true-label rank. Since each term in L is $< \hat{p}(y^*)$ and $|\mathcal{B}| \leq K - 1$, $L \leq (K-1)\hat{p}(y^*)$, so $s = 1 - L$ yields (4) after substituting (A1)–(A2). Taking expectations and applying (A3) gives (5); differentiating gives monotonicity in C when $\varepsilon < \bar{h}$. Rearranging (4) gives the threshold form in (6), and differentiating $T(C)$ shows the coverage upper bound decreases with concentration under the theorem condition.

Conservative bound verification. Using $\varepsilon = 0$ and $\bar{h} = 1/K$, Theorem 1(ii) yields conservative lower bounds verified on all five applicable tasks: s-shipcond (0.518 vs. observed 0.98), s-payterms (0.545 vs. 0.92), s-group (0.474 vs. 1.00), i-plant (0.261 vs. 0.86), and s-incoterms (0.296 vs. 0.87). The large gaps between bounds and observed values reflect conservative assumptions in the $(K-1)$ counting bound and the uniform residual prior $\bar{h} = 1/K$; tighter bounds follow from empirical $\bar{h}$ estimates.

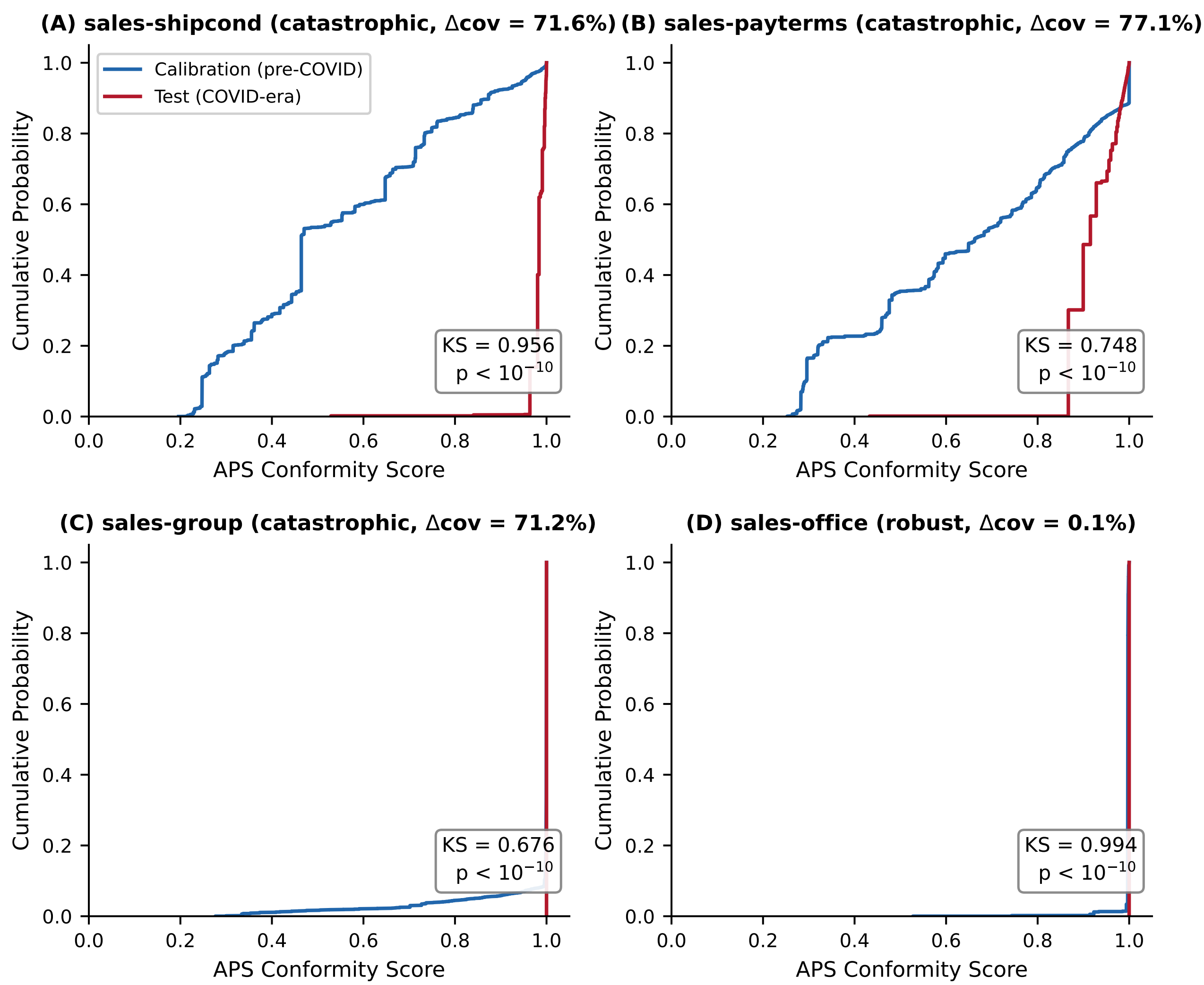

Figure 3: **Conformity score CDFs (seed=42).** Catastrophic tasks show rightward test-score shifts (stochastic dominance), while robust tasks can show near-degenerate or left-shift behavior.

I CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 11 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 11: KS Tests: Calibration vs. Test Conformity Score Distributions (seed=42).

Task	KS	p	Mean_{cal}	$\text{Mean}_{\text{test}}$
Catastrophic tasks (test scores > calibration)				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
Robust tasks				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Non-randomized APS scores (conservative coverage guarantee). For catastrophic tasks, test scores shift rightward (stochastic dominance), consistent with Theorem 1. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

J RAPS COMPARISON

Table 12 compares APS and RAPS coverage drops across all 8 tasks with 10-seed means ($\lambda = 0.01$, $k_{\text{reg}} = 5$).

Table 12: APS vs. RAPS Coverage Drop (10-seed means $\pm$ std). $\lambda = 0.01$, $k_{\text{reg}} = 5$.

Task	Cl	APS Drop	RAPS Drop	Δ	Note
Severe tasks					
s-shipcond	45	60.4 ± 31.8	67.8 ± 25.4	+7.4	No improvement
s-group	462	73.5 ± 33.9	10.4 ± 1.3	-63.1	RAPS protects
s-payterms	135	79.9 ± 29.4	35.1 ± 28.6	-44.8	Partial recovery
Non-severe tasks					
i-plant	35	11.8 ± 9.0	13.1 ± 5.8	+1.3	Near-identical
i-shippoint	70	9.3 ± 27.6	20.5 ± 21.6	+11.2	RAPS worse
i-incoterms	13	10.8 ± 9.1	10.8 ± 9.1	0.0	Identical
s-incoterms	13	6.7 ± 6.8	6.7 ± 6.8	0.0	Identical
s-office	25	0.04 ± 0.03	0.04 ± 0.03	0.0	Identical

All values are 10-seed means $\pm$ std. APS drops differ from 50-seed Table 1 due to smaller seed range (10 vs. 50) and high variance. The key comparison is APS vs. RAPS within the same seed set. RAPS eliminates the catastrophic drop for sales-group by truncating the class accumulation that amplifies score inflation in many-class settings (mean set size: 232/462 classes). Sales-shipcond is unhelped because single-feature dependence places dominant mass on the wrong class *before* the $k_{\text{reg}} = 5$ penalty activates—the exchangeability violation occurs at the top of the APS ordering, not through class accumulation. For low-class tasks ($\leq$35 classes), RAPS and APS produce identical or near-identical drops, confirming RAPS regularization has negligible effect when prediction sets are already small. Item-shippoint ($K = 70$) is an exception: RAPS worsens coverage by 11.2 pp (paired Wilcoxon $p = 0.064$, 10 seeds), reflecting that the regularization penalty can disrupt calibration for intermediate-cardinality tasks with high seed variance.

K MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with `RandomForestClassifier` (500 trees), `XGBClassifier`, and `CatBoostClassifier` (ordered boosting), all with seed=42 and the same data pipeline (Table 13).

Table 13: SHAP Concentration and Coverage Drop Across Model Classes.

Task	LGB	RF	XGB	CB	LGB	RF	XGB	CB
	Concentration (%)				Coverage Drop (%)			
s-shipcond	50.7	57.4	44.6	53.7	71.6	15.8	47.2	1.1
s-group	47.3	22.5	33.3	31.8	71.2	50.0	56.8	14.5
s-payterms	54.2	27.9	30.6	25.7	77.1	13.8	20.4	−5.3
i-shippoint	48.8	52.1	41.0	43.1	18.5	22.7	80.4	61.8
i-plant	23.9	26.8	26.0	29.0	10.6	15.3	9.7	8.5
s-incoterms	23.7	26.8	21.6	28.2	8.5	2.5	3.7	−1.5
i-incoterms	28.9	17.5	18.4	21.7	11.3	2.9	1.1	−0.4
s-office	42.6	24.5	36.8	33.5	0.1	0.0	0.0	−0.0

Results. The diagnostic's predictive power varies by model class (Table 14). All three gradient-boosted implementations show positive correlation: LightGBM ($\rho = 0.833$, $p = 0.010$), CatBoost ($\rho = 0.667$, $p = 0.071$), and XGBoost ($\rho = 0.548$, $p = 0.16$). RandomForest ($\rho = 0.30$, $p = 0.47$) does not replicate.

Mixed-effects analysis. Since (task, model) pairs share data and temporal split, we fit a linear mixed-effects model with task as a random intercept to properly account for clustering: $\text{Drop}_{ij} = \beta_0 + \beta_1 \cdot \text{Concentration}_{ij} + u_i + \varepsilon_{ij}$, where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the task random effect. Across the three gradient-boosted models ($n = 24$, 8 tasks $\times$ 3 models), the concentration effect is $\beta_1 = 1.64$ (95% CI $[0.70, 2.58]$, $p = 0.0006$): each 1% increase in concentration predicts 1.6 pp additional coverage drop, after accounting for task-level clustering. We note that with only 8 task-level clusters the random-intercept variance may be downward-biased and the fixed-effect standard error potentially anti-conservative; nonetheless, the large effect size ($\beta_1 = 1.64$) and narrow CI suggest the direction is robust. Including model class as a fixed effect yields $\beta_1 = 1.66$ ($p = 0.001$). With all four models including RF ($n = 32$): $\beta_1 = 1.06$ ($p = 0.010$).

Table 14: Concentration–Drop Relationship by Model Class.

Model	Method	Conc.–Drop ρ	p
LightGBM	Boosting (GOSS)	**0.833**	**0.010**
CatBoost	Boosting (ordered)	0.667	0.071
XGBoost	Boosting (histogram)	0.548	0.16
RandomForest	Bagging	0.30	0.47
Mixed-effects, 3 boosting ($n = 24$)		$\beta_1 = 1.64$	<**0.001**
Mixed-effects, all 4 ($n = 32$)		$\beta_1 = 1.06$	**0.010**

Interpretation. All three gradient-boosted implementations show the concentration–drop relationship, but with different vulnerability profiles. CatBoost's ordered boosting produces dramatically different task-level vulnerabilities: LGB-catastrophic tasks (s-shipcond: 71.6% LGB drop) show only 1.1% CatBoost drop, while i-shippoint reverses (18.5% LGB $\rightarrow$ 61.8% CatBoost). This confirms each model's SHAP concentration captures its *own* learned dependence—not a universal data property—and that cross-model transfer is unreliable (LGB conc. $\rightarrow$ CatBoost drop: $\rho = 0.07$). RF's bagging produces smoother probability surfaces, compressing both concentration range and vulnerability profiles. Specifically, RF's failure to replicate ($\rho = 0.30$) has two mechanistic causes: (1) *compressed concentration range*—RF concentrations span 17.5–57.4% vs. LGB's 23.7–54.2%, but the RF range is driven by a single outlier (s-shipcond at 57.4%); excluding it, the RF range is 17.5–26.8%, offering almost no discriminative signal; (2) *smoothed probability surfaces*—bagging averages 500 independent trees, each seeing random feature subsets, which distributes effective dependence more evenly than boosting's sequential error-correction. The result is that RF rarely develops the concentrated single-feature dependence that drives catastrophic conformal failure in boosting models, and consequently SHAP concentration has limited predictive power for RF vulnerability.

The diagnostic is *model-specific*: it measures how a particular model's learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.

Neural network extension. To test beyond tree-based models, we evaluate a 2-layer MLP (128/64 hidden units, ReLU, softmax, permutation importance) on all 8 SALT tasks (10 seeds for sales-level tasks, 3 for item-level). The within-MLP concentration–drop correlation is $\rho = 0.43$ ($p = 0.29$), similar to RF and not significant. However, the model-specificity claim is strikingly confirmed across all 8 tasks: the MLP learns fundamentally different concentration profiles from tree-based models. Key examples: (1) i-shippoint shows MLP $C = 77.2\%$, drop $= 82.3$ pp vs. LGB $C = 48.8\%$, drop $= 18.5$ pp—the MLP concentrates nearly all importance on a single feature and fails catastrophically, while LGB distributes importance and maintains coverage; (2) s-incoterms shows MLP $C = 30.3\%$, drop $= 42.3$ pp vs. LGB $C = 23.7\%$, drop $= 8.5$ pp—the MLP's greater concentration predicts its greater failure; (3) s-office shows MLP $C \approx 0\%$, drop $= 0.1$ pp—perfect robustness prediction. The non-significant within-MLP ρ is driven by two tasks (s-group: $C = 27.5\%$, drop $= 78.4$ pp; s-payterms: $C = 28.0\%$, drop $= 60.6$ pp) where the MLP distributes importance evenly yet still fails catastrophically—suggesting the MLP's failure mechanism involves global sensitivity to the shift rather than concentrated single-feature dependence. This confirms that SHAP concentration is diagnostic specifically for the concentrated-dependence failure mode that characterizes gradient-boosted models, not a universal predictor of all failure modes.